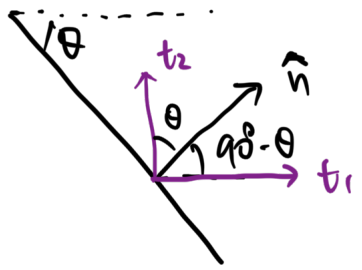
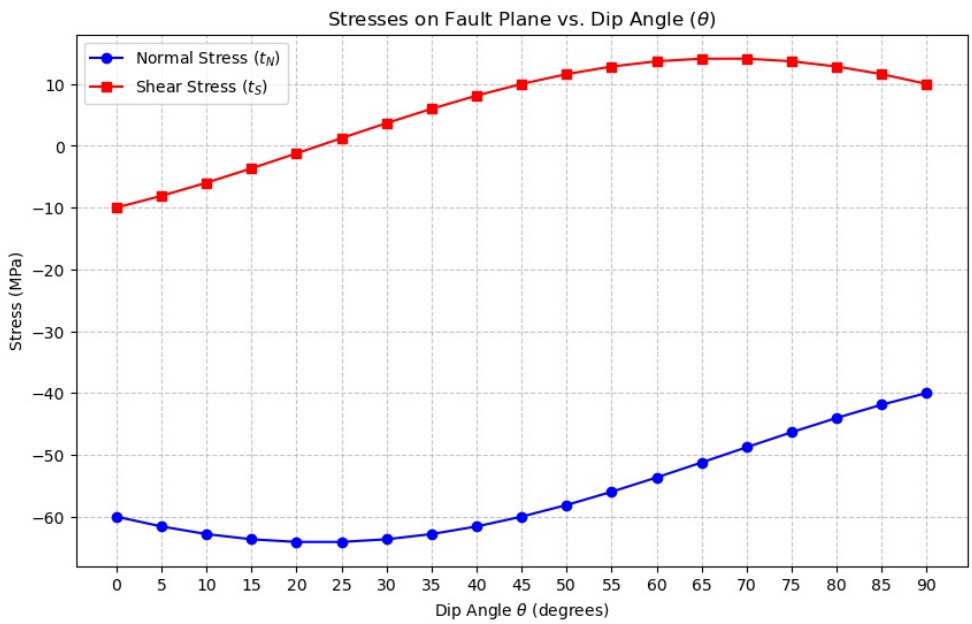
HW 2

(1)



$$\hat{n} = \begin{pmatrix} \cos(90^\circ-\theta) \\ \sin(90^\circ-\theta) \end{pmatrix} = \begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix}$$



斜截面应力法

$$\therefore t(\hat{n}) = \tau\hat{n} = \begin{pmatrix} -40 & -10 \\ -10 & -60 \end{pmatrix}\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} = \begin{pmatrix} -40\sin\theta - 10\cos\theta \\ -10\sin\theta - 60\cos\theta \end{pmatrix}$$

$$\therefore t_n = t(\hat{n}) \cdot \begin{pmatrix} \cos(90^\circ-\theta) \\ \cos\theta \end{pmatrix} = -50 - 10\cos(2\theta) - 10\sin(2\theta) \quad (\text{MPa})$$

$$t_s = t(\hat{n}) \cdot \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix} = 10\sin(2\theta) - 10\cos(2\theta) \quad (\text{MPa})$$

旋转法

逆时针旋转矩阵 $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$，旋转 $\alpha = 90^\circ - \theta$，使 x轴与 $\vec{n}$ 重合

$$\sigma' = R\sigma R^T = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} -40 & -10 \\ -10 & -60 \end{bmatrix}\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} -50 - 10\cos(2\theta) - 10\sin(2\theta) & 10\cos(2\theta) - 10\sin(2\theta) \\ 10\cos(2\theta) - 10\sin(2\theta) & -50 + 10\cos(2\theta) + 10\sin(2\theta) \end{bmatrix}$$

$$\therefore T(\vec{n}) = \sigma' \cdot (1,0)^T = \Big(\underbrace{-50 - 10\cos(2\theta) - 10\sin(2\theta)}_{\text{正应力}},\ \underbrace{10\cos(2\theta) - 10\sin(2\theta)}_{\text{-剪切应力}}\Big)^T$$

法向面力

(2) $\begin{vmatrix} -40-\lambda & -10 \\ -10 & -60-\lambda \end{vmatrix} = (-40-\lambda)(-60-\lambda)-(10)^2=0 \Rightarrow \lambda = -50 \pm 10\sqrt{2}$

$\therefore$ 最大主应力 $\sigma_1 = -50+10\sqrt{2}$ (MPa)

最小主应力 $\sigma_2 = -50-10\sqrt{2}$ (MPa)

考虑旋转阵与任意 σ

$$\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \sigma_x & \sigma_{xy} \\ \sigma_{yx} & \sigma_y \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos(2\alpha) + \sigma_{xy}\sin(2\alpha) & \cdots \\ \frac{\sigma_x-\sigma_y}{2}\sin(2\alpha) - \sigma_{xy}\cos(2\alpha) & \cdots \end{bmatrix}$$

$\therefore$ 可知正应力 σ_α 为 $\frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos(2\alpha) + \sigma_{xy}\sin(2\alpha)$

剪应力 τ_α 为 $-\frac{\sigma_x-\sigma_y}{2}\sin(2\theta) + \sigma_{xy}\cos(2\alpha)$

最大正应力的剪应力为0的方向，令 $\tau_\alpha=0$，可得 $\tan(2\alpha) = \frac{2\tau_{xy}}{\sigma_x-\sigma_y}$

代入本题数据 $\tan(2\alpha)=-1$ $\therefore 2\alpha = -45^\circ$ 或 135°

$\therefore \alpha_1 = -22.5^\circ$，对应 σ_1

$\alpha_2 = 67.5^\circ$，对应 σ_2

(3) 最大剪应力

以最大主应力方向为x轴方向，此时 $\sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$

$\therefore \tau_\alpha = -\frac{\sigma_1-\sigma_2}{2}\sin(2\alpha)$

可知当 $2\alpha = \pm\frac{\pi}{2}$ 时，$|\tau_\alpha|$ 取极大，此时 $2\alpha = \pm\frac{\pi}{4}$

$\therefore \tau_{max} = \frac{\sigma_1-\sigma_2}{2} = 10\sqrt{2}$ (MPa)

方向分别为 $\alpha_1 = -22.5^\circ + 45^\circ = 22.5^\circ$

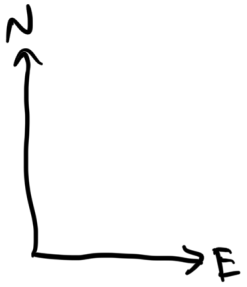
$\alpha_2 = -22.5^\circ - 45^\circ = -67.5^\circ$

(1)

$$\alpha^2 = \frac{\lambda + 2\mu}{\rho} \qquad \beta^2 = \frac{\mu}{\rho} \Rightarrow \begin{cases} \lambda + 2\mu = \rho\alpha^2 \\ \mu = \rho\beta^2 \end{cases} \Rightarrow \begin{cases} \lambda = 3.105 \times 10^{10} \\ \mu = 3.3075 \times 10^{10} \end{cases}$$

$$1\,Pa = 1\,kg/(m \cdot s^2)$$

(2)



$$e = \begin{bmatrix} -0.26 & -0.69 \\ -0.69 & 0.92 \end{bmatrix} \times 10^{-6}$$

$$\sigma_{ij} = \lambda\theta\delta_{ij} + 2\mu e_{ij} \qquad \theta = e_{11} + e_{22} = 0.66 \times 10^{-6}$$

$$\therefore \sigma = \begin{bmatrix} \lambda\theta + 2\mu e_{11} & 2\mu e_{12} \\ 2\mu e_{21} & \lambda\theta + 2\mu e_{22} \end{bmatrix} = \begin{bmatrix} 3294 & -45643.5 \\ -45643.5 & 8135 \end{bmatrix} (Pa$$

(3)

$$\tan 2\theta = \frac{2e_{12}}{e_{11} - e_{22}} \approx 1.1695 \Rightarrow \begin{cases} \theta_1 = 24.74^\circ \\ \theta_2 = 114.74^\circ \end{cases} \xrightarrow[\text{相对N}]{\text{转换}} \begin{cases} \text{Azimuth}_1 = 65.26^\circ \\ \text{Azimuth}_2 = 155.26^\circ \end{cases}$$

(4)

$$E_{11} = (0.101 \times 10^{-6}/yr) \times 1000\,yr = 1.01 \times 10^{-4}$$

$$E_{22} = (-0.02 \times 10^{-6}/yr) \times 1000\,yr = -0.2 \times 10^{-4}$$

$$E_{12} = (0.005 \times 10^{-6}/yr) \times 1000\,yr = 0.05 \times 10^{-4}$$

$$E_{kk} = 0.81 \times 10^{-4}$$

$$\therefore \tau = \begin{bmatrix} \lambda E_{kk} + 2\mu E_{11} & 2\mu E_{12} \\ 2\mu E_{12} & \lambda E_{kk} + 2\mu E_{22} \end{bmatrix} = \begin{bmatrix} 9.2 & 0.33 \\ 0.33 & 1.19 \end{bmatrix} \times 10^6\ Pa$$

(5)

$$\frac{\Delta A}{A_0} \doteq e_{11} + e_{22}$$

$$\Delta A_{year} = A_0 \times E_{kk} = 0.081\,m^2$$

$$\Delta A_{Earthquake} = A_0 \times \theta = 0.66\,m^2$$

第一题编程

计算结果按照倾角 5 度为间隔编程画图展示

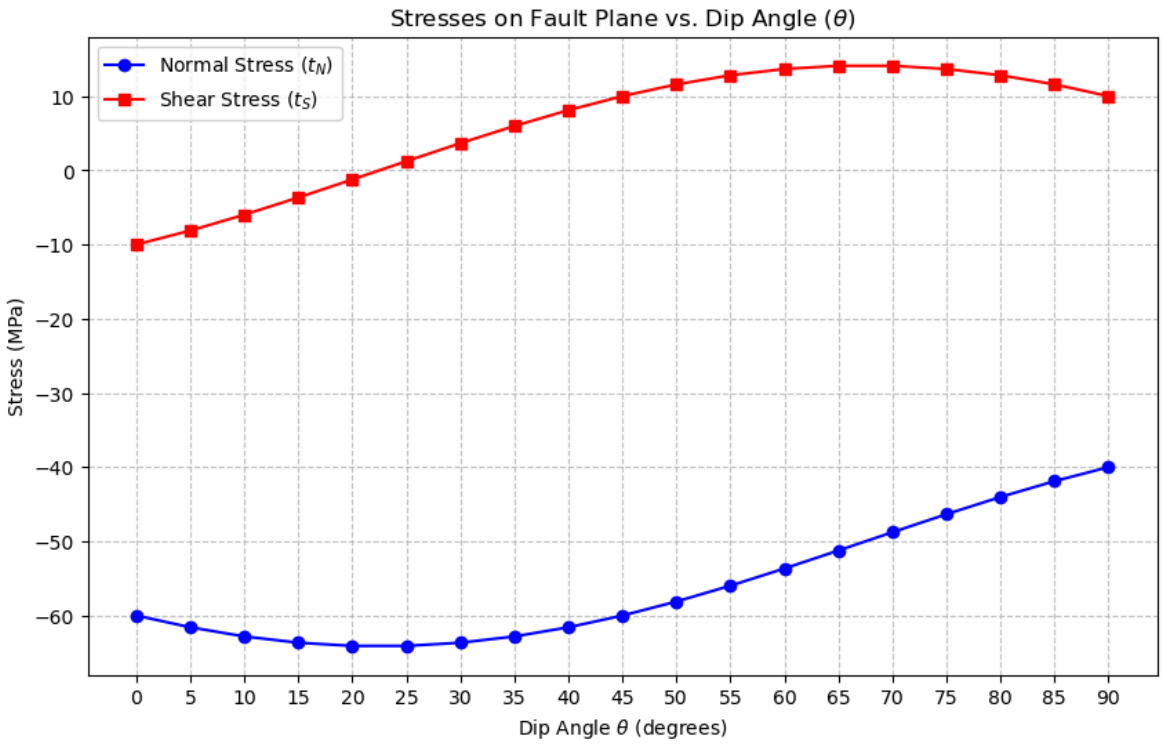
In [8]:
```
import numpy as np
import matplotlib.pyplot as plt

# 设置角度范围 0 到 90 度，间隔 5 度
thetas_deg = np.arange(0, 95, 5)
thetas_rad = np.radians(thetas_deg)

# 计算正应力 sigma_N 和 剪应力 t_S
sigma_N = -50 - 10 * np.cos(2 * thetas_rad) - 10 * np.sin(2 * thetas_rad)
t_S = 10 * np.sin(2 * thetas_rad) - 10 * np.cos(2 * thetas_rad)

# 绘图
plt.figure(figsize=(10, 6))
plt.plot(thetas_deg, sigma_N, 'bo-', label='Normal Stress ($t_N$)')
plt.plot(thetas_deg, t_S, 'rs-', label='Shear Stress ($t_S$)')

plt.title('Stresses on Fault Plane vs. Dip Angle ($\\theta$)')
plt.xlabel('Dip Angle $\\theta$ (degrees)')
plt.ylabel('Stress (MPa)')
plt.xticks(np.arange(0, 95, 5))
plt.grid(True, linestyle='--', alpha=0.7)
plt.legend()
plt.show()
```



第二题编程

(f)

正应力的计算公式为:

$$\tau_n = \tau_{11}\cos^2\phi - 2\tau_{12}\sin\phi\cos\phi + \tau_{22}\sin^2\phi$$

剪应力：

$$\tau_s = (\tau_{11} - \tau_{22})\sin\phi\cos\phi + \tau_{12}(\cos^2\phi - \sin^2\phi)$$

In [ ]:
```
import numpy as np
import pandas as pd

landers = {'t11': 3294, 't22': 81351, 't12': -45643.5}

longterm = {'t11': 9200385, 't22': 1192035, 't12': 330750}

def calculate_fault_stresses(t11, t22, t12, azimuth_deg):
    """计算特定方位角直立断层上的正应力和剪应力"""
    phi = np.radians(azimuth_deg)


    tau_n = 0.5 * (t11 + t22) + 0.5 * (t11 - t22) * np.cos(2 * phi) - t12
    tau_s = 0.5 * (t11 - t22) * np.sin(2 * phi) + t12 * np.cos(2 * phi)
    return tau_n, tau_s

results = []

for az in range(0, 180, 10):
    n_L, s_L = calculate_fault_stresses(landers['t11'], landers['t22'], l
    n_LT, s_LT = calculate_fault_stresses(longterm['t11'], longterm['t22'

    results.append({
        'Azimuth (deg)': az,
        'Landers Normal (kPa)': n_L / 1000,
        'Landers Shear (kPa)': s_L / 1000,
        'Long-term Normal (MPa)': n_LT / 1000000,
        'Long-term Shear (MPa)': s_LT / 1000000
    })

df = pd.DataFrame(results)
print(df.round(2).to_string(index=False))
```

```
 Azimuth (deg)  Landers Normal (kPa)  Landers Shear (kPa)  Long-term Norma
l (MPa)  Long-term Shear (MPa)
             0                  3.29               -45.64
9.20                    0.33
            10                 21.26               -56.24
8.85                    1.68
            20                 41.76               -60.05
8.05                    2.83
            30                 62.34               -56.62
6.91                    3.63
            40                 80.50               -46.36
5.57                    4.00
            50                 94.05               -30.51
4.18                    3.89
            60                101.37               -10.98
2.91                    3.30
            70                101.56                 9.88
1.92                    2.32
            80                 94.61                29.54
1.32                    1.06
            90                 81.35                45.64
1.19                   -0.33
           100                 63.39                56.24
1.55                   -1.68
           110                 42.88                60.05
2.34                   -2.83
           120                 22.31                56.62
3.48                   -3.63
           130                  4.15                46.36
4.83                   -4.00
           140                 -9.40                30.51
6.22                   -3.89
           150                -16.72                10.98
7.48                   -3.30
           160                -16.91                -9.88
8.48                   -2.32
           170                 -9.96               -29.54
9.07                   -1.06
```

对于 Landers 地震同震应力变化： 最大剪应力出现在方位角 20° 和 110°（绝对值约为 60.05 kPa）。

对于 1000 年长期累积应力： 最大剪应力出现在方位角 40° 和 130°（绝对值约为 4.00 MPa）

(g)

In [7]:

```python
import numpy as np
import pandas as pd

yearly_stress = {'t11': 9.20, 't22': 1.19, 't12': 0.33}
mu_s = 0.2

def calculate_yearly_cff(t11, t22, t12, azimuth_deg, mu):

    phi = np.radians(azimuth_deg)
```

```
    tau_n = 0.5 * (t11 + t22) + 0.5 * (t11 - t22) * np.cos(2 * phi) - t12
    tau_s = 0.5 * (t11 - t22) * np.sin(2 * phi) + t12 * np.cos(2 * phi)


    delta_cff = abs(tau_s) + mu * tau_n
    return tau_n, tau_s, delta_cff

results = []
for az in range(0, 180, 10):
    n, s, cff = calculate_yearly_cff(yearly_stress['t11'], yearly_stress[

    results.append({
        'Azimuth (deg)': az,
        'Yearly ΔNormal (kPa/yr)': n,
        'Yearly ΔShear (kPa/yr)': s,
        'Yearly ΔCFF (kPa/yr)': cff
    })
df = pd.DataFrame(results)
print(df.round(2).to_string(index=False))
```

```
 Azimuth (deg)  Yearly ΔNormal (kPa/yr)  Yearly ΔShear (kPa/yr)  Yearly ΔC
FF (kPa/yr)
             0                     9.20                    0.33
2.17
            10                     8.85                    1.68
3.45
            20                     8.05                    2.83
4.44
            30                     6.91                    3.63
5.02
            40                     5.57                    4.00
5.11
            50                     4.17                    3.89
4.72
            60                     2.91                    3.30
3.88
            70                     1.91                    2.32
2.70
            80                     1.32                    1.06
1.32
            90                     1.19                   -0.33
0.57
           100                     1.54                   -1.68
1.99
           110                     2.34                   -2.83
3.29
           120                     3.48                   -3.63
4.33
           130                     4.82                   -4.00
4.97
           140                     6.22                   -3.89
5.13
           150                     7.48                   -3.30
4.80
           160                     8.48                   -2.32
4.02
           170                     9.07                   -1.06
2.87
```

(h)

In [ ]:
```
import numpy as np
import pandas as pd

landers = {'t11': 3.29, 't22': 81.35, 't12': -45.64}
longterm = {'t11': 9200.39, 't22': 1192.04, 't12': 330.75}
yearly = {'t11': 9.20, 't22': 1.19, 't12': 0.33}
mu_s = 0.2

def calc_stresses(t11, t22, t12, azimuth_deg):
    phi = np.radians(azimuth_deg)
    tau_n = 0.5 * (t11 + t22) + 0.5 * (t11 - t22) * np.cos(2 * phi) - t12
    tau_s = 0.5 * (t11 - t22) * np.sin(2 * phi) + t12 * np.cos(2 * phi)
    return tau_n, tau_s

results = []

for az in range(0, 180, 10):
    n_L, s_L = calc_stresses(landers['t11'], landers['t22'], landers['t12
    n_1000, s_1000 = calc_stresses(longterm['t11'], longterm['t22'], long
    n_a, s_a = calc_stresses(yearly['t11'], yearly['t22'], yearly['t12'],

    cff_a = abs(s_a) + mu_s * n_a

    delta_cff_instant = (abs(s_1000 + s_L) - abs(s_1000)) + mu_s * n_L

    dt_years = delta_cff_instant / cff_a

    sign_agree = (s_1000 * s_L) > 0

    results.append({
        'Azimuth': az,
        'Sign Agree?': 'Yes' if sign_agree else 'No',
        'Instant ΔCFF (kPa)': delta_cff_instant,
        'Δt (Years)': dt_years
    })

df = pd.DataFrame(results)
print(df.round(2).to_string(index=False))
```

Azimuth	Sign Agree?	Instant ΔCFF (kPa)	Δt (Years)
0	No	-44.98	-20.73
10	No	-51.99	-15.07
20	No	-51.70	-11.65
30	No	-44.15	-8.80
40	No	-30.26	-5.92
50	No	-11.70	-2.48
60	No	9.29	2.39
70	Yes	30.19	11.16
80	Yes	48.46	36.62
90	No	-29.37	-51.71
100	No	-43.56	-21.90
110	No	-51.47	-15.62
120	No	-52.16	-12.05
130	No	-45.53	-9.17
140	No	-32.39	-6.31
150	No	-14.33	-2.98
160	Yes	6.49	1.62
170	Yes	27.55	9.58

(i)

PFO 附近没有出现地震活动高峰，仅仅意味着当地没有那些既处于 CFF 增加区，又恰好处于临界破裂边缘的断层。这并不能作为否定 CFF 阈值模型的证据；相反，它是断层力学中非常典型且合理的现象。